

Bit-masking
01

88 → logical and
8 → Bit-and
11 → logical or
1 → Bit-or

SAT
101
101
000

[2, 3, 2, 3, 2]

$\overline{A \times B}$

$\sim(1) = 0$

	a	b	a & b	a b	a ^ b
	1	1	1	1	0
	0	1	0	1	1
	1	0	0	1	1
	0	0	0	0	0

$\overline{a \wedge b} = a$

$a \wedge b \wedge c = b \wedge c \wedge a \wedge b$

$a \wedge a = 0$



if (n < 2 == 0) {

000 000 101
0001

$\overline{n \& 1}$

$1 - (n \& 1) = 1 - 1 = 0$

Left shift (←)

$\times 0000101 \rightarrow 5$

$5 \ll 2 \rightarrow 5 \times 2^2 \rightarrow 20$

$5 \ll 3 \rightarrow 5 \times 2^3$

$a \ll n \rightarrow a \times 2^n$

$00001010 \rightarrow 10$ (5 < 1)

$0001010 \leftarrow 20$

(5 < 2)

Right shift

$n_1 = 22$

$n_1 \gg 2 \rightarrow \frac{22}{2^2} = \frac{22}{4} = 5$

$a_1 \gg n = \left\lfloor \frac{a_1}{2^n} \right\rfloor$

00010011
11101100
11101101 → -15

$\pm 1110110^x$

1111011
00000100
101

(-5)

8 00010000 → mask
01010100
43210

01010100
43210

$n = (n \gg 1)$

$(n \& 1) == 0$

(1 < 4)

10 < 4
100 < 4
1000 < 4
10000

00000101
81

A magic number is defined as a number which can be expressed as a power of 5 or sum of unique powers of 5. First few magic numbers are 5, 25, 30(5 + 25), 125, 130(125 + 5),

Write a function to find the nth Magic number.

Example:

Input: n = 2

Output: 25

Input: n = 5

Output: 130

1	2	3	4	5	6	7	8	9	10	11	12
5	25	30	125	130	150	155	625	630	650	655	750
5^1	5^2	$5^1 + 5^2$	5^3	$5^3 + 5^1$	$5^3 + 5^2$	$5^3 + 5^2 + 5^1$	5^4	$5^4 + 5^1$	$5^4 + 5^2$	$5^4 + 5^2 + 5^1$	$5^4 + 5^3$

$i = 8$

mul = 1

while (i > 0) {
 rem = i % 2;
 sum = sum + rem * mul;
 mul = mul * 5;
 i = i / 2;
}

$i = 8$

110
 $5^3 5^2 5^1$

while (i > 0) {
 if ((i & 1) != 0) {
 sum = sum + mul;
 mul = mul * 5;
 i >> 1;
 }

4 7 44 47 74 77 444 447 474 477 744 747 774 777
1 2 3 4 5 6 7 8 9 10

$2 + 4 + 11$

num

474744

1 ✓
2 ✓
3 ✓
4 ✓
5 ✓
20

$2 + 2^2 + 2^3 + \dots + 2^{n-1}$

$\frac{2(2^n - 1)}{2 - 1} = 2^n - 2$
 $\frac{2(2^n - 1)}{2 - 1} = 2^n - 2$

474774
47474
4744
44

$2^6 - 2 = 62$

$2^1 + 2^2 + 2^3 + 2^4 = 22$

$1 \ll 1 + 1 \ll 2 + 1 \ll 4$

$84 + 1 = 85$

abc	pos
0 1 2	8
0	000 → -
1	001 → a
2	010 → b
3	011 → ab
4	100 → c
5	101 → ac
6	110 → bc
7	111 → abc

```
public static void print(String s) {  
    int n = s.length();  
    for (int i = 0; i < (1 << n); i++) {  
        Subsequence(s, i);  
    }  
}
```

$N = s.length() \rightarrow 3$

pos = 0

```
private static void Subsequence(String s, int i) {  
    // TODO Auto-generated method stub  
    String ans = "";  
    int pos = 0;  
    while (i > 0) {  
        if ((i & 1) != 0) {  
            ans = ans + s.charAt(pos);  
        }  
        pos++;  
        i >>= 1;  
    }  
    System.out.println(ans);  
}
```

$i = 5$

101
 10
 1
 $1 \rightarrow 10$

ans = ac

pos = 7, 2, 4, 6, 8, 10

$i = 8$

101
 10
 1
 $1 \rightarrow 10$

ans = ac

00011100
11100011
11100100
00011100
1100

[5, 6, 31, 14, 6, 14, 5, 6, 14, 31, 2]

5 6 31 14 6 14 5 6 14 31 2

5 6 31 14 6 14 5 6 14 31 2

5 6 31 14 6 14 5 6 14 31 2

5 6 31 14 6 14 5 6 14 31 2

0000100

mask

0101 0100
0100 0